

Python Fundamentals (Assignment4)

Assignment Problems

| Concept: Classes & Objects

Q1. Create a `BankAccount` class with **attributes** `account_number`, `owner_name`, and `balance`.

Add **methods** to `deposit`, `withdraw`, and `check balance`.

| Concept: Classes & Objects

Q2. Create a class `Book` with the following attributes:

- title
- author
- list of reviews

And add methods to:

- add a new review
- count reviews
- display all reviews

| Concept: Encapsulation

Q3. Create a class `Student` with **private** attributes `_name`, `_roll_no`, and `_marks`. Provide **getter** and **setter** methods with validation (e.g., marks cannot be negative, roll number has to be between 1 & 100 & name cannot be empty).

| Concept: Function Overriding

Q4. Create a class `Shape` with a method `area()`.

Create subclasses `Circle`, `Rectangle`, and `Triangle` that **override** the `area()` method.

Concept: Inheritance

Q5. Create a **base** class **Vehicle** with attributes like brand and model. Create two **subclasses** **Car** and **Bike** that add extra attributes - seats (in Car) & engine_cc (in Bike).

Concept: Abstraction

Q6. Create an **abstract** class **Employee** with an **abstract method** `calculate_salary()`. Create subclasses **Intern**, **FullTimeEmployee**, and **ContractEmployee** that implement the method differently.

Concept: Constructor Overloading (with Default Parameters)

Q7. Create a class **Person** that allows the constructor to work with:

- name only
- name + age
- name + age + address

As direct constructor overloading (multiple constructors) are not allowed but we have to use **default parameters** to simulate constructor overloading.

Concept: Instance & Class Attributes

Q8. Create a class **Player** with:

- a class variable `player_count`
- instance variables `name` and `level`

Track how many players were created.

Concept: Multiple Inheritance

Q9. Create the following classes: **Herbivore**, **Carnivore**, **Omnivore** with some attributes & methods. Then create a class **Bear** that inherits from all the above classes to showcase how multiple inheritance works.

Concept: OOP

Q10. Mini Project – OOP Chat System

Let's create a Chat System using OOPs concepts. We have to create classes:

- User
- Message
- ChatRoom

And we have to implement functions:

- sending messages
- viewing chat history
- user joining and leaving the chatroom